

MATHS PORTFOLIO



TOPIC:

Pair of Linear Equations
in Two Variables

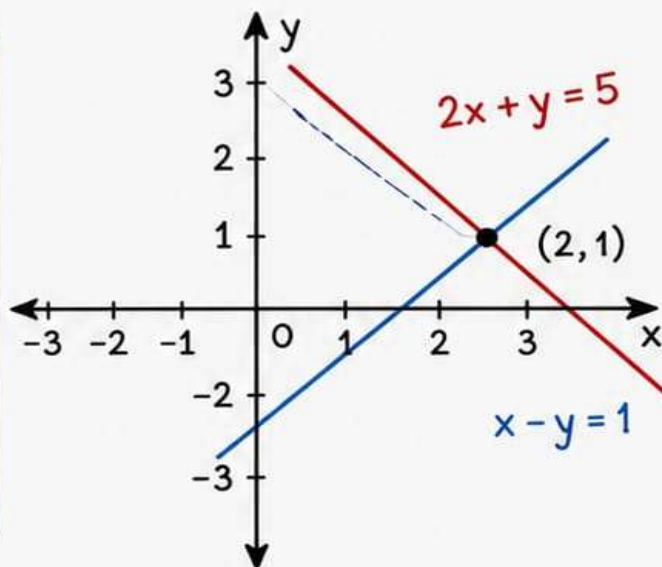
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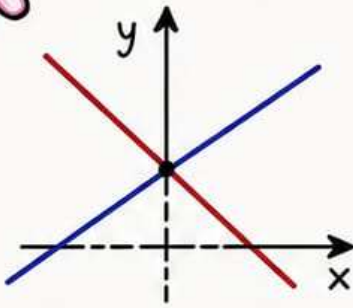
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“Mathematics is the language
of the universe.”



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INDEX

$$2x + y = 5$$
$$x - y = 1$$



Topic: Pair of Linear Equations in Two Variables



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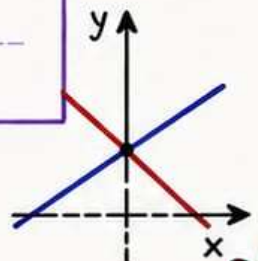
y



π



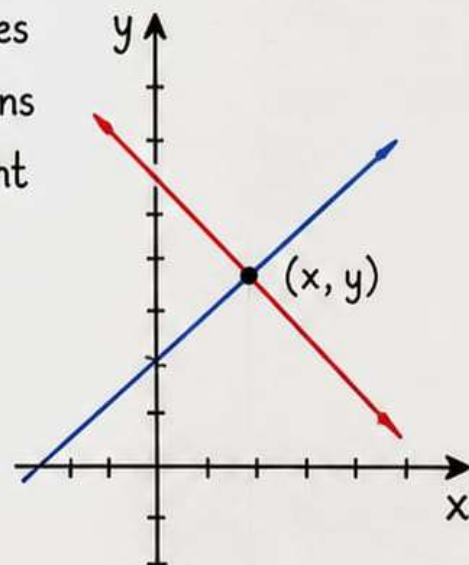
Learn. Solve. Understand.
Math makes sense!





INTRODUCTION

A pair of linear equations in two variables consists of two equations with two unknowns (x and y). These equations represent straight lines when drawn on a graph.



The general form is:

$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

These equations can be solved using different methods like graphical method, substitution method and elimination method.

Solution is the ordered pair (x, y) which satisfies both equations.

They are useful in solving real-life problems involving two unknown quantities such as cost, distance, speed, time, mixtures, profit and loss etc.

Example :

Cost of 2 apples and 1 mango = ₹30

Cost of 1 apple and 2 mangoes = ₹40

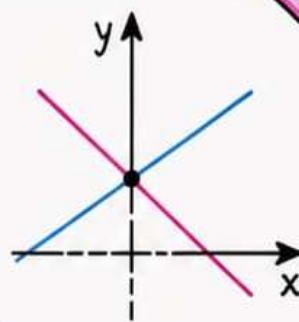


These can be written as linear equations and solved.





REAL-LIFE APPLICATIONS



Pair of linear equations in two variables are used to solve many real-life problems in different situations.



They are used in:

1

Business and Commerce

To find cost price and selling price, profit and loss, number of items sold, total cost, etc.



For example, finding the cost of two items using two given conditions.

2

Distance, Speed and Time

To find speed, distance or time when two conditions are given.



For example, a car covers a distance at two different speeds or times.

3

Mixture Problems

To find the quantity of two ingredients in a mixture.



For example, mixing milk and water in a certain ratio.

4

Budget and Finance

To manage income and expenses in a budget.



For example, distributing money in two different categories.

5

Compare Two Situations

To compare two quantities or conditions.



For example, comparing the costs of two different options.



Thus, pair of linear equations in two variables helps us solve many practical problems of everyday life.

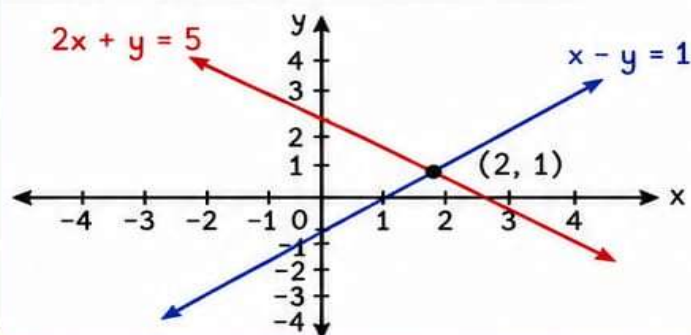


GRAPHICAL REPRESENTATION

Representation of all possible cases for a pair of linear equations in two variables

1. INTERSECTING LINES

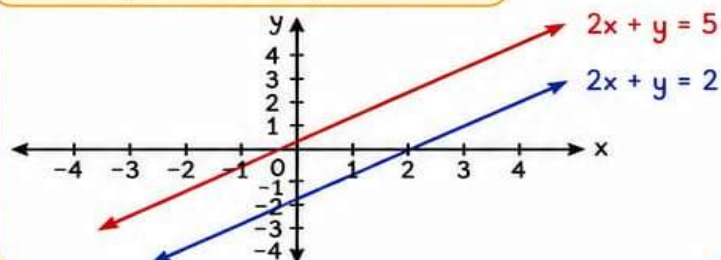
(One unique solution)



- The two lines intersect at one point.
- Solution $(x, y) = (2, 1)$
- Consistent and Independent

2. PARALLEL LINES

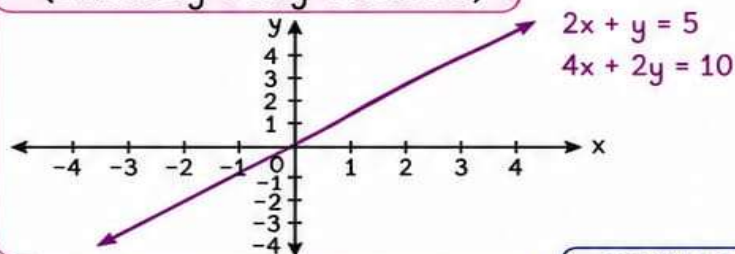
(No solution)



- The two lines are parallel.
- They never intersect.
- No solution.
- Inconsistent

3. COINCIDENT LINES

(Infinitely many solutions)



- The two lines coincide.
- They overlap each other.
- Infinitely many solutions.
- Consistent and Dependent

SUMMARY

Case	Intersecting Lines	Parallel Lines	Coincident Lines
Graph			
Condition	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
Number of Solutions	One (Unique)	No Solution	Infinitely Many
Type	Consistent (Independent)	Inconsistent	Consistent (Dependent)

NOTE :

- * Intersecting lines meet at one point.
- * Parallel lines never meet.
- * Coincident lines are the same line.





EXAMPLE PROBLEM (SOLVED)

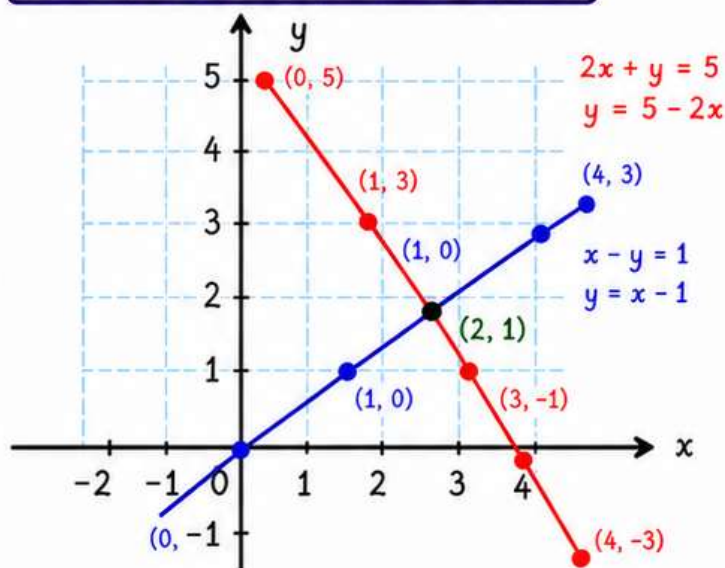


Problem:

Solve the pair of linear equations:

$$\begin{cases} 2x + y = 5 \dots(1) \\ x - y = 1 \dots(2) \end{cases}$$

METHOD 1: GRAPHICAL METHOD



- Draw both lines using the points from the table.
- The point where they intersect is the solution.
- From the graph, the lines meet at:

Points to Plot

For $2x + y = 5$ ($y = 5 - 2x$)			For $x - y = 1$ ($y = x - 1$)		
x	y	(x, y)	x	y	(x, y)
0	5	(0, 5)	0	-1	(0, -1)
1	3	(1, 3)	1	0	(1, 0)
2	1	(2, 1)	2	1	(2, 1)
3	-1	(3, -1)	3	2	(3, 2)
4	-3	(4, -3)	4	3	(4, 3)

Solution: $(x, y) = (2, 1)$

METHOD 2: SUBSTITUTION METHOD

Step 1: From equation (2) $x - y = 1$

$$x = y + 1$$

Step 2: Substitute into equation (1) $2x + y = 5$

$$2(y + 1) + y = 5$$

$$2y + 2 + y = 5$$

$$3y + 2 = 5$$

$$3y = 3$$

$$y = 1$$

Step 3: Substitute $y = 1$ in $x = y + 1$

$$x = 1 + 1$$

$$x = 2$$

Solution: $(x, y) = (2, 1)$

METHOD 3: ELIMINATION METHOD

$$\begin{cases} 2x + y = 5 \dots(1) \\ x - y = 1 \dots(2) \end{cases}$$

Step 1: Add both equations

$$(2x + y) + (x - y) = 5 + 1$$
$$3x = 6$$

Step 2: $x = \frac{6}{3} = 2$

$$x = 2$$

Step 3: Substitute $x = 2$ in equation (2)

$$2 - y = 1$$

$$-y = 1 - 2 = -1$$

$$y = 1$$

Solution: $(x, y) = (2, 1)$

FINAL ANSWER:

$$x = 2, y = 1$$



Verification:

Substitute in equation (1):

$$2(2) + 1 = 5$$

$$4 + 1 = 5$$





GRAPHICAL SOLUTION

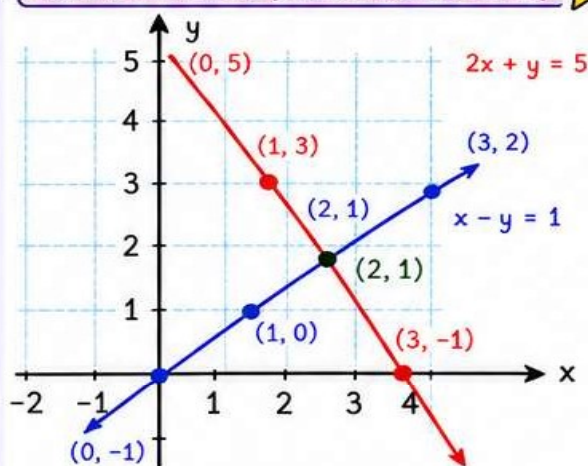


A graphical solution of a pair of linear equations is obtained by plotting both equations on a graph. The point where the two lines intersect represents the solution.

EXAMPLE PAIR OF EQUATIONS:

$$\begin{aligned} 2x + y &= 5 \\ x - y &= 1 \end{aligned}$$

EXAMPLE GRAPH (INTERSECTING LINES)



STEPS TO SOLVE GRAPHICALLY:

Step 1: Convert to slope-intercept form $\begin{cases} y = 5 - 2x \\ y = x - 1 \end{cases}$

Step 2: Plot points

For $y = 5 - 2x$
($2x + y = 5$)

x	y	(x, y)
0	5	(0, 5)
1	3	(1, 3)
2	1	(2, 1)
3	-1	(3, -1)

For $y = x - 1$
($x - y = 1$)

x	y	(x, y)
0	-1	(0, -1)
1	0	(1, 0)
2	1	(2, 1)
3	2	(3, 2)

Step 3: Draw the lines

- Plot the points on graph paper
- Join each set of points to form two straight lines

Step 4: Find the intersection point
The lines intersect at (2, 1)

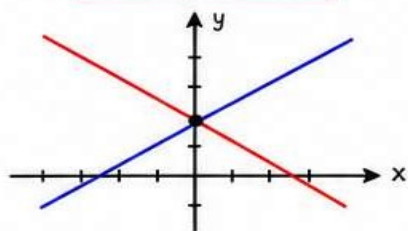
FINAL ANSWER: $x = 2, y = 1$

★ GRAPHICAL SOLUTION: ALL POSSIBILITIES ★

1. INTERSECTING LINES

Lines intersect at one point.

One unique solution



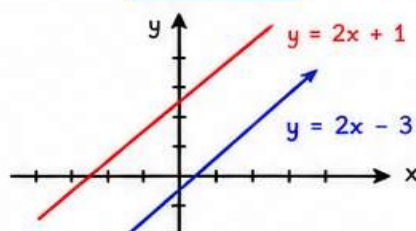
Example: $\begin{cases} 2x + y = 5 \\ x - y = 1 \end{cases}$

Solution: (2, 1)

2. PARALLEL LINES

Lines are parallel and never meet.

No solution



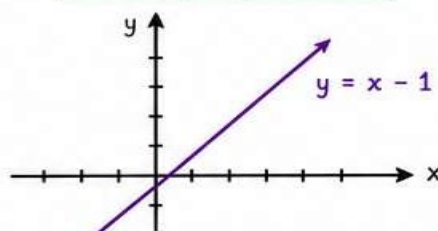
Example: $\begin{cases} y = 2x + 1 \\ y = 2x - 3 \end{cases}$

Solution: No solution

3. COINCIDENT LINES

Lines lie on top of each other.

Infinitely many solutions



Example: $\begin{cases} y = x - 1 \\ y = x - 1 \end{cases}$

Solution: Infinitely many solutions

CONCLUSION:

- ✓ Intersecting lines → One unique solution
- ✓ Parallel lines → No solution
- ✓ Coincident lines → Infinitely many solutions

"The beauty of mathematics is that every problem has a solution - you just need the right method to find it!"



KEY CONCEPTS

Key concepts help us understand the nature of solutions for a pair of linear equations in two variables.

1. GENERAL FORM

A pair of linear equations in two variables can be written as:

$$a_1x + b_1y + c_1 = 0 \quad \text{and} \quad a_2x + b_2y + c_2 = 0$$

where $a_1, b_1, c_1, a_2, b_2, c_2$ are real numbers and a_1, b_1, a_2, b_2 are not both zero.

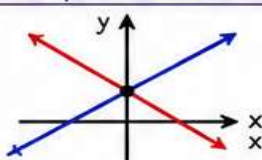
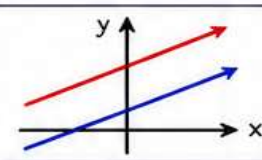
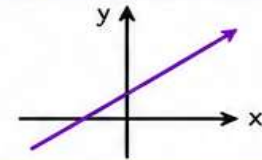
2. SLOPE OF A LINE

A linear equation in the form $y = mx + c$ has slope m .

If a line is written as $ax + by + c = 0$, then its slope is

$$m = -\frac{a}{b} \quad (\text{where } b \neq 0)$$

3. NATURE OF LINES AND SOLUTIONS

Condition	Slopes	Intercepts (y-intercepts)	Graphical Representation	Number of Solutions
Intersecting Lines	$m_1 \neq m_2$ or $-\frac{a_1}{b_1} \neq -\frac{a_2}{b_2}$	$c_1 \neq c_2$ (or lines cut each other)		One unique solution
Parallel Lines	$m_1 = m_2$ (or $-\frac{a_1}{b_1} = -\frac{a_2}{b_2}$)	$c_1 \neq c_2$ (or lines do not meet)		No solution
Coincident Lines	$m_1 = m_2$ and $c_1 = c_2$ (or $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$)	$c_1 = c_2$ (or lines lie on each other)		Infinitely many solutions

4. CONSISTENCY OF A PAIR OF LINEAR EQUATIONS

- If the lines intersect, the pair is **consistent** and has a **unique solution**.
- If the lines are parallel, the pair is **inconsistent** and has **no solution**.
- If the lines are coincident, the pair is **dependent** and has **infinitely many solutions**.

5. KEY TAKEAWAYS

- ★ The graph of a pair of linear equations reveals the number of solutions.
- ★ Slopes and intercepts help us determine the nature of lines.
- ★ Understanding the relationship between coefficients helps in identifying solutions.

"The more you understand the concepts, the easier the solutions become!"

ACTIVITY (REAL-LIFE PROBLEMS)

We use pairs of linear equations in our daily life to solve many real-world situations. Let's see how!



1. AGE PROBLEM

The sum of the ages of a father and his son is 50 years. After 10 years, the father's age will be twice his son's age. Find their present ages.

Solution:

Let father's present age = x years

Let son's present age = y years



Forming Equations:

$$x + y = 50 \quad \dots (1)$$

$$x + 10 = 2(y + 10) \quad \dots (2)$$

Solving:

From (2): $x + 10 = 2y + 20$

$$x - 2y = 10 \quad \dots (3)$$

From (1): $x + y = 50 \quad \dots (1)$

Subtract (3) from (1):

$$(x + y) - (x - 2y) = 50 - 10$$

$$x + y - x + 2y = 40$$

$$3y = 40 \Rightarrow y = \frac{40}{3} = 13\frac{1}{3}$$

Substitute in (1):

$$x + 13\frac{1}{3} = 50$$

$$x = 50 - 13\frac{1}{3} = 36\frac{2}{3}$$

Answer:

Father's present age = $36\frac{2}{3}$ years

Son's present age = $13\frac{1}{3}$ years

★ Check:

$$\text{Sum} = 36\frac{2}{3} + 13\frac{1}{3} = 50 \quad \checkmark$$

After 10 years:

$$\text{Father} = 46\frac{2}{3}, \text{ Son} = 23\frac{1}{3}$$

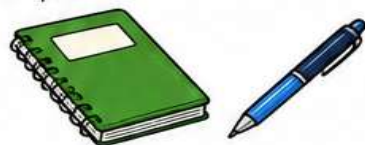
$$46\frac{2}{3} = 2 \times 23\frac{1}{3} \quad \checkmark$$

2. COST PROBLEM

The cost of 3 notebooks and 2 pens is ₹140.

The cost of 2 notebooks and 5 pens is ₹190.

Find the cost of one notebook and one pen.



Solution:

Let cost of one notebook = ₹ x

Let cost of one pen = ₹ y

Forming Equations:

$$3x + 2y = 140 \quad \dots (1)$$

$$2x + 5y = 190 \quad \dots (2)$$

Solving:

Multiply (1) by 5 and (2) by 2:

$$15x + 10y = 700 \quad \dots (3)$$

$$4x + 10y = 380 \quad \dots (4)$$

Subtract (4) from (3):

$$11x = 320 \Rightarrow x = \frac{320}{11} = 29.09$$

Substitute in (1):

$$3(29.09) + 2y = 140$$

$$87.27 + 2y = 140$$

$$2y = 52.73 \Rightarrow y = 26.36$$

Answer:

Cost of one notebook = ₹29.09

Cost of one pen = ₹26.36

★ Check:

$$3(29.09) + 2(26.36) = 87.27 + 52.72 = 140 \quad \checkmark$$

$$2(29.09) + 5(26.36) = 58.18 + 131.80 = 189.98 \approx 190 \quad \checkmark$$

3. DISTANCE AND SPEED PROBLEM

A car covers a certain distance at a speed of 60 km/h and takes 2 hours less than the time taken to cover the same distance at a speed of 40 km/h. Find the distance and the time taken at each speed.



Solution:

Let distance = d km

Time taken at 60 km/h = x hours

Time taken at 40 km/h = y hours

Forming Equations:

$$60x = d \quad \dots (1)$$

$$40y = d \quad \dots (2)$$

$$\text{Also, } y - x = 2 \quad \dots (3)$$

Solving:

From (1) and (2): $60x = 40y$

$$3x = 2y \Rightarrow y = \frac{3x}{2} \quad \dots (4)$$

Substitute in (3):

$$\frac{3x}{2} - x = 2$$

$$\frac{3x - 2x}{2} = 2 \Rightarrow \frac{x}{2} = 2 \Rightarrow x = 4$$

From (4): $y = \frac{3(4)}{2} = 6$

Distance $d = 60x = 60 \times 4 = 240$ km

Answer:

Distance = 240 km

Time at 60 km/h = 4 hours

Time at 40 km/h = 6 hours

★ Check:

At 60 km/h: Time = $\frac{240}{60} = 4$ hours $\checkmark$

At 40 km/h: Time = $\frac{240}{40} = 6$ hours $\checkmark$

Difference = $6 - 4 = 2$ hours $\checkmark$

Key Takeaway:

Real-life problems can be solved easily by forming two linear equations in two variables and solving them using any suitable method.

"Mathematics makes life simple when we know how to apply it!"



IMPORTANT FORMULAS

1. STANDARD FORM

The general form of a linear equation in two variables is:

$$ax + by + c = 0$$

where a , b , c are real numbers and a and b are not both zero.

2. SLOPE-INTERCEPT FORM

A linear equation can be written as:

$$y = mx + c$$

- m = slope of the line
- c = y -intercept (value of y when $x = 0$)

3. SLOPE OF A LINE

If a line is written in the form $ax + by + c = 0$, then the slope (m) of the line is:

$$m = -\frac{a}{b} \quad (\text{where } b \neq 0)$$

4. INTERCEPTS

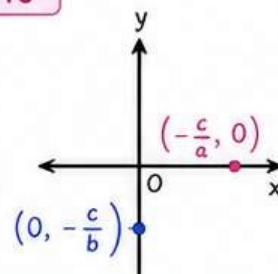
If $ax + by + c = 0$

- x -intercept (when $y = 0$):

$$x = -\frac{c}{a} \quad (a \neq 0)$$

- y -intercept (when $x = 0$):

$$y = -\frac{c}{b} \quad (b \neq 0)$$



5. CONDITIONS FOR NATURE OF LINES

Condition	In Terms of Slopes	In Terms of Coefficients For equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$	Graphical Representation	Number of Solutions
Intersecting Lines	$m_1 \neq m_2$ (or) $-\frac{a_1}{b_1} \neq -\frac{a_2}{b_2}$	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$		One unique solution
Parallel Lines	$m_1 = m_2$ (or) $-\frac{a_1}{b_1} = -\frac{a_2}{b_2}$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$		No solution
Coincident Lines	$m_1 = m_2$ and $c_1 = c_2$ (or) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$		Infinitely many solutions

6. SOLVING FORMULAS (Algebraic Method)

For equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

$$x = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \quad y = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

(provided $a_1b_2 - a_2b_1 \neq 0$)

7. CROSS MULTIPLICATION METHOD (For Proportional Case)

$$\text{If } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \longrightarrow$$

Then the lines are coincident and have infinitely many solutions.

"Formulas are the keys that unlock the door to every solution."

CONCLUSION

$$\begin{aligned}2x + y &= 5 \\ x - y &= 1\end{aligned}$$

Key Takeaways

- ✓ A pair of linear equations in two variables has two equations with two variables.
- ✓ Graphical method helps us find the solution by the point of intersection of the two lines.
- ✓ Substitution and Elimination methods help us find the solution algebraically.
- ✓ The solution represents values of the variables that satisfy both equations.
- ✓ These concepts help us solve real-life problems easily and accurately.

What We Learned

- We understood the meaning and types of solutions of a pair of linear equations:
 - ★ Unique Solution – Lines intersect at one point.
 - ★ No Solution – Lines are parallel.
 - ★ Infinitely Many Solutions – Lines are coincident.
- We learned different methods to solve the equations.
- We applied our knowledge to solve problems and interpret the results.
- We saw how mathematics is useful in our daily lives.

Why It Matters

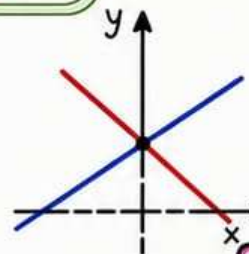


Solving pairs of linear equations helps us make smart decisions, plan resources, compare quantities and understand real-world situations better.



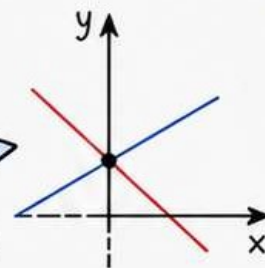
In the End

We explored, practiced and discovered that mathematics is not just about numbers and equations—it is about understanding the world around us and finding solutions that make life better!





THANK YOU!



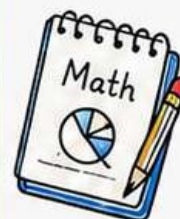
Dear Teachers, Mentors & Evaluators,



I would like to extend my sincere gratitude to all the teachers and guides who supported and encouraged me throughout the completion of this Mathematics Portfolio.



- ★ Thank you for providing the knowledge, guidance and resources that helped me understand the concepts deeply.
- ★ I truly appreciate your *patience*, *motivation* and *valuable feedback*, which inspired me to strive for excellence.
- ★ This portfolio reflects not just my learning, but also the support and encouragement I received from all of you.



I am grateful for the opportunity to learn, explore and present my understanding in a well-structured and meaningful way.
Thank you for being a part of my *academic journey*!



By

Name : Vaira Lingam
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Subject : Mathematics
School : Krishnamal Ramasubbaiyer School
Academic Year : 2026 - 2027
Motto : *Learn Today, Lead Tomorrow.*



“ The beautiful thing about learning is that no one can take it away from you.
Thank you for helping me build knowledge, confidence and a love for mathematics.
With sincere thanks,
Vaira Lingam ”

